

BMW Market Analytics API Documentation

COMP3011 – Web Services and Web Data

1. Overview

This document describes all endpoints exposed by the BMW Market Analytics API.

Base URL (local development):

<http://127.0.0.1:8000>

All responses are returned in JSON format.

2. Authentication

Write operations require an API key.

Header:

X-API-Key:

Example:

X-API-Key: super-secret-key

If the key is missing or invalid:

Status Code: 401 Unauthorized

Response:

```
{  
  "detail": "Invalid or missing API key"  
}
```

Read-only endpoints do not require authentication.

3. Endpoints

3.1 Health

GET /health

Description:

Returns API status.

Authentication: Not required

Example Request:

GET /health

Example Response (200 OK):

```
{  
  "status": "ok"  
}
```

3.2 Car Models

GET /models

Description:

Returns all car models.

Authentication: Not required

Example Request:

GET /models

Example Response:

```
[  
  {  
    "id": 1,  
    "name": "3 Series",  
    "series": "3 Series",  
    "body_style": null,  
    "fuel_type": "Petrol",  
    "transmission": "Automatic"  
  }  
]
```

GET /models/{id}

Path Parameter:

- id(integer)

Description:

Returns a single model by ID.

Authentication: Not required

Example Request:

GET /models/1

Example Response:

```
{
  "id": 1,
  "name": "3 Series",
  "series": "3 Series",
  "body_style": null,
  "fuel_type": "Petrol",
  "transmission": "Automatic"
}
```

Error (404 Not Found):

```
{
  "detail": "Model not found"
}
```

POST /models

Authentication: Required

Request Body:

```
{
  "name": "X5",
  "series": "X5",
  "body_style": "SUV",
  "fuel_type": "Diesel",
  "transmission": "Automatic"
}
```

Example Request:

POST /models

Example Response (201 Created):

```
{  
  "id": 10,  
  "name": "X5",  
  "series": "X5",  
  "body_style": "SUV",  
  "fuel_type": "Diesel",  
  "transmission": "Automatic"  
}
```

PATCH /models/{id}

Authentication: Required

Path Parameter:

- id (integer)

Request Body Example:

```
{  
  "fuel_type": "Hybrid"  
}
```

Example Request:

PATCH /models/1

Example Response:

```
{  
  "id": 1,  
  "name": "3 Series",  
  "series": "3 Series",  
  "body_style": null,  
  "fuel_type": "Hybrid",  
  "transmission": "Automatic"  
}
```

DELETE /models/{id}

Authentication: Required

Example Request:

DELETE /models/1

Success Response:

Status Code: 204 No Content

3.3 Market Records

GET /models/{id}/records

Path Parameter:

- id(integer)

Description:

Returns all price records for a model.

Authentication: Not required

Example Request:

GET /models/1/records

Example Response:

```
[
{
  "id": 20,
  "car_model_id": 1,
  "year": 2018,
  "price": 19000.0
}
```

POST /models/{id}/records

Authentication: Required

Path Parameter:

- id(integer)

Request Body:

```
{  
  "year": 2018,  
  "price": 19000  
}
```

Example Request:

POST /models/1/records

Example Response (201 Created):

```
{  
  "id": 25,  
  "car_model_id": 1,  
  "year": 2018,  
  "price": 19000.0  
}
```

Error (404 Not Found):

```
{  
  "detail": "Model not found"  
}
```

3.4 Analytics Endpoints

GET /analytics/average-price

Query Parameter:

- model_id (integer, required)

Example Request:

GET /analytics/average-price?model_id=1

Example Response:

```
{  
  "model_id": 1,  
  "model_name": "3 Series",  
  "average_price": 18250.75  
}
```

GET /analytics/price-trend

Query Parameter:

- model_id (integer, required)

Example Request:

GET /analytics/price-trend?model_id=1

Example Response:

```
{
  "model_id": 1,
  "model_name": "3 Series",
  "trend": [
    {
      "year": 2016,
      "average_price": 18000.0,
      "num_records": 12
    }
  ]
}
```

GET /analytics/top-models

Query Parameters:

- year (integer, required)
- limit (integer, optional)

Example Request:

GET /analytics/top-models?year=2018&limit=5

Example Response:

```
{
  "year": 2018,
  "results": [
    {
      "model_id": 3,
      "model_name": "X5",
      "average_price": 35000.0,
      "num_records": 7
    }
  ]
}
```

```
]
}
```

4. Error Codes Summary

Code	Meaning
200	Successful request
201	Resource created
204	Resource deleted
401	Invalid or missing API key
404	Resource not found

5. Setup Notes

- Run database migrations before starting the API.
- Import dataset using: `python -m scripts.import_bmw`
- Start server with: `uvicorn app.main:app --reload`